Experiment 2: The Aggregation Effect

Federated Learning Grokking Project

March 2026

1 Experiment Design

Central question. Each FL client holds only n/K samples—far below the centralised phase boundary $\alpha_c \approx 0.198$ established in Exp 1. A centralised model trained on the same n/K samples never groks. Does FedAvg aggregation recover the grokking ability of the full dataset?

Three conditions per cell. For every (α, K) pair we run three matched experiments (3 seeds each):

1. **Centralized-full**: standard centralised training on $n_{\text{train}} = \alpha \cdot p^2$ samples. This is the *ceiling*—the best any method can do with this data.
2. **Centralized-reduced**: centralised training on n_{train}/K samples. This is the *floor*—what a single client could achieve alone.
3. **FL IID**: K clients, each holding n_{train}/K IID samples, trained with FedAvg (5 local epochs, full participation).

Grid.

- $\alpha \in \{0.20, 0.25, 0.30, 0.35, 0.50\}$
- $K \in \{2, 5, 10, 20, 50, 97\}$
- 30 cells $\times$ 3 conditions $\times$ 3 seeds = **270 training runs**
- Budget: $S_{\max} = 50{,}000$ gradient steps (FL) and 100,000 steps (centralised)
- All other hyperparameters as Exp 1: $N = 256$, $p = 97$, GD with $\text{lr} = 50$, MSE loss

What "aggregation rescues" means. If FL groks but the centralized-reduced baseline does not, the *only* difference is that FedAvg averages K model updates. Each client trains on data that is individually insufficient for grokking; aggregation pools the gradient signal without pooling the data. This is the core mechanism we aim to isolate.

2 Phase Diagram: Where Does Aggregation Matter?

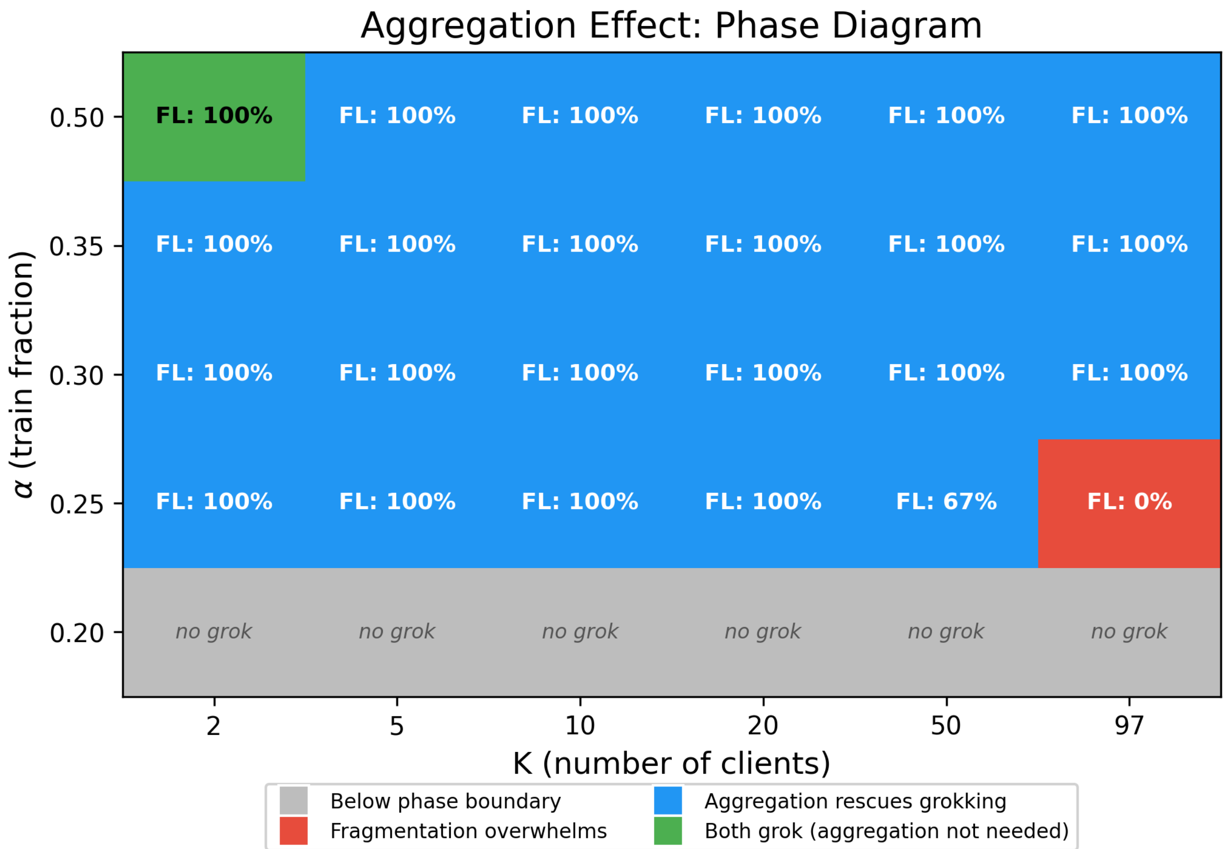

Figure 1: Phase diagram in (α, K) space. Each cell is coloured by outcome category: **gray** = below the centralized phase boundary ($\alpha = 0.2$), **green** = both FL and centralized-reduced grok, blue = FL groks but centralized-reduced does not (aggregation rescues), **red** = neither FL nor centralized-reduced groks (fragmentation overwhelms). Annotations show the FL grokking success rate.

Key observations.

- **Aggregation rescues grokking across the vast majority of the grid.** The dominant colour is blue: FL groks with 100% seed success even though a centralised model trained on the same per-client data (α/K) never groks. This is the experiment's headline result.
- **Only one cell shows "both grok":** $\alpha = 0.5, K = 2$ ($\alpha_{\text{eff}} = 0.25$ per client). This is the only configuration where per-client data alone exceeds the phase boundary. Every other cell with $K \geq 5$ has $\alpha_{\text{eff}} < 0.198$, meaning individual clients are provably below the grokking threshold.
- **Fragmentation overwhelms only at the extreme.** FL fails at ($\alpha = 0.25, K = 97$) where each client sees just 24 samples ($\alpha_{\text{eff}} = 0.003$), and partially at ($\alpha = 0.25, K = 50$) where 2/3 seeds grok. These are the only red/partial cells in the entire grid.
- $\alpha = 0.2$ **is below the centralized boundary** and serves as a negative control: neither centralized-full nor FL groks at any K, confirming that aggregation cannot manufacture

grokking from fundamentally insufficient total data.

3 Quantifying the Slowdown

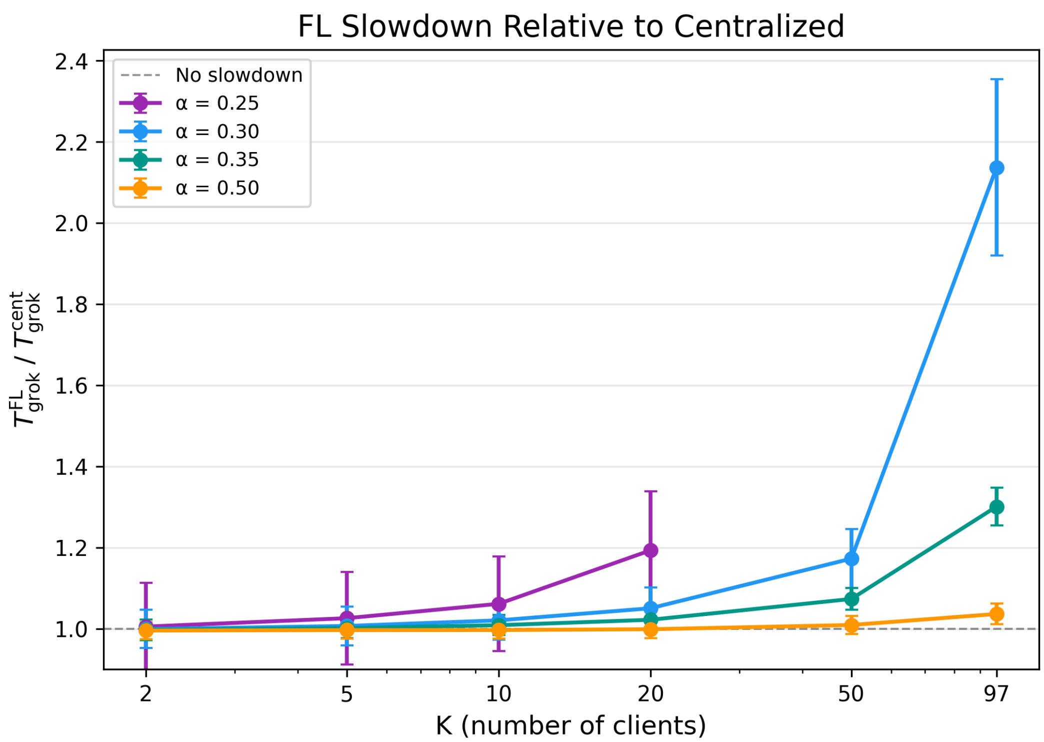

Figure 2: Slowdown ratio $T^{\mathrm{FL}}_{\mathrm{grok}}/T^{\mathrm{cent}}_{\mathrm{grok}}$ vs. K. A ratio of 1.0 means FL is as fast as centralised. Lines stop where FL fails to grok.

Key observations.

- **For $\alpha = 0.5$, FL has essentially zero overhead.** The ratio stays below 1.04 even at $K = 97$. With abundant data, FedAvg adds no measurable cost—the global model converges as if trained centrally.
- **The slowdown grows with K and is worse near the phase boundary.** At $\alpha = 0.3$, the ratio rises gently to 1.17 at $K = 50$ then jumps to 2.14 at $K = 97$. At $\alpha = 0.25$ (closest to the boundary), the ratio reaches 1.19 at $K = 20$ before FL fails entirely at $K \geq 50$.
- **Regime structure:**
  - $K \leq 20$: slowdown $< 1.2\times$ across all α that grok
  - $K = 50$: moderate slowdown (1.01–1.17×)
  - $K = 97$: significant slowdown (1.04–2.14×), and FL breaks at low α
- **Interpretation.** The slowdown is a competition between aggregation benefit and client drift. At small K, each client has enough data that 5 local epochs do not cause significant drift. At large K, each client overfits to its tiny shard, producing updates that partially cancel upon averaging. The fact that FL still groks even at $K = 97$ (for $\alpha \geq 0.3$) shows that aggregation is remarkably robust.

4 FL vs Centralized: Scatter View

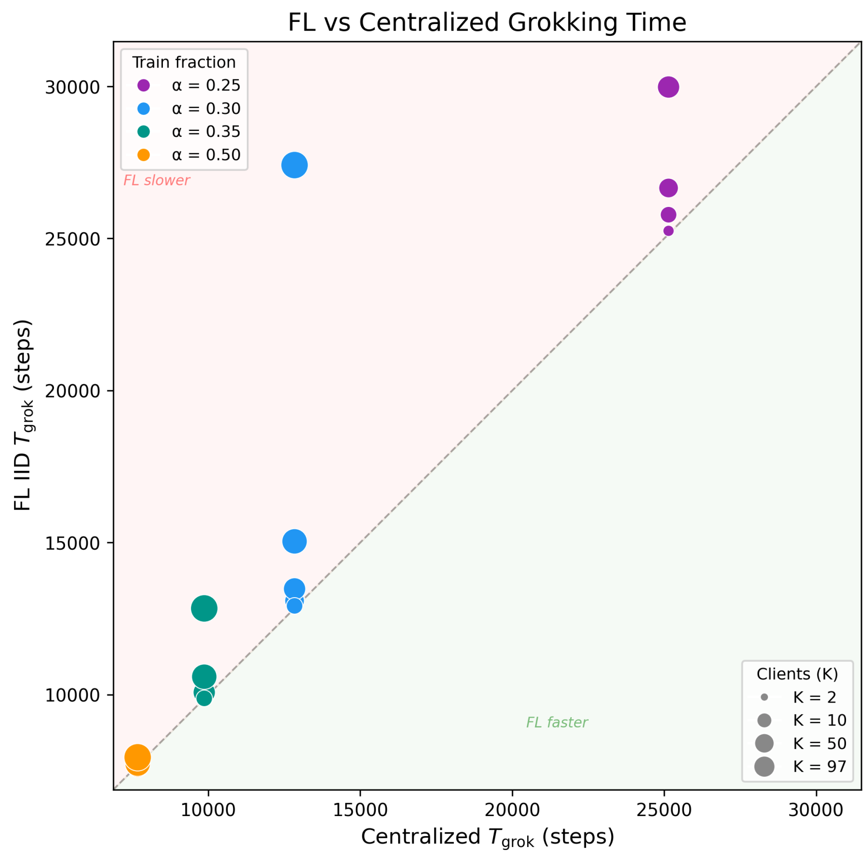

Figure 3: FL grokking time vs. centralized grokking time. Marker colour encodes α; marker size encodes K. Points above the diagonal indicate FL is slower than centralized.

Key observations.

- **Small K points cluster tightly on the diagonal**, confirming that FL with few clients is indistinguishable from centralized training in terms of grokking speed.
- **Large K pulls points above the diagonal**, especially for lower α. The $\alpha = 0.3, K = 97$ point (large blue marker) is the most extreme: FL takes $\sim$ 27,400 steps vs. $\sim$ 12,800 centralized.
- $\alpha = 0.25$ **(purple) points are the most scattered** and closest to failure, reflecting proximity to the phase boundary.

5 Representative Training Dynamics

Three regimes, three panels.

- **Panel (a):** $\alpha = 0.50, K = 2$ — **all grok.** All three conditions reach 100% test accuracy. Centralized-full and FL overlap almost perfectly ($T_{grok} \approx$ 7,600). Centralized-reduced is 3.3× slower ($T_{grok} \approx$ 25,100) because it trains on only half the data. This is the only cell where the per-client data ($\alpha_{eff} = 0.25$) exceeds the phase boundary.

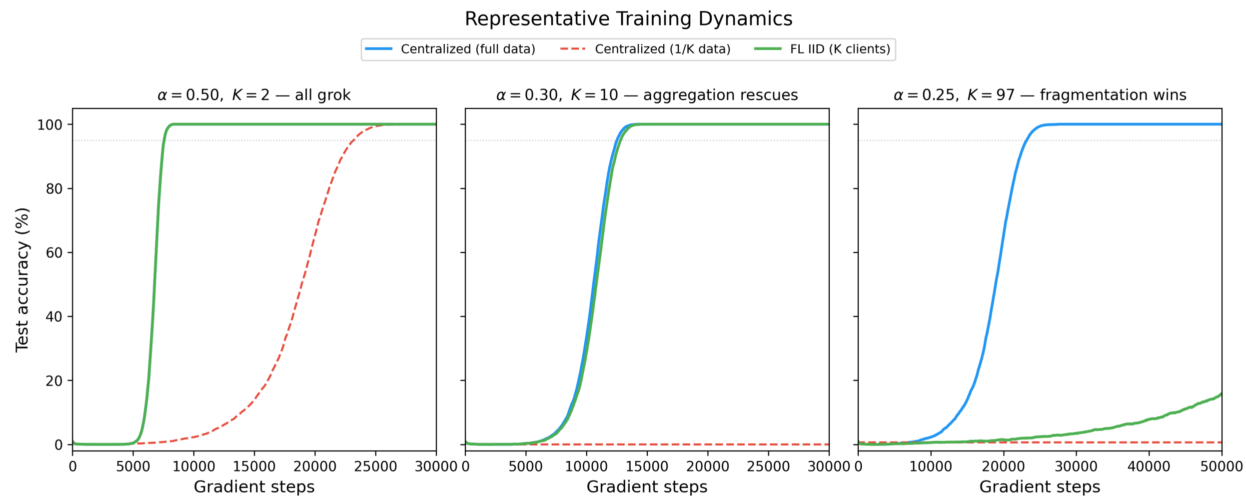

Figure 4: Test accuracy over time for three representative cells, illustrating the three outcome regimes. Blue: centralized-full, red dashed: centralized-reduced, green: FL IID.

- **Panel (b):** $\alpha = 0.30, K = 10$ — **aggregation rescues.** Centralized-full groks at $\sim 12{,}800$ steps. FL follows close behind at $\sim 13{,}100$ steps. Centralized-reduced, training on $\alpha_{\text{eff}} = 0.03$, is permanently stuck at 0% test accuracy. This is the cleanest demonstration: 10 clients with individually hopeless data shards collectively recover the grokking solution through aggregation alone.
- **Panel (c):** $\alpha = 0.25, K = 97$ — **fragmentation wins.** Centralized-full groks (slowly, at $\sim 25{,}100$ steps), but FL with 97 clients fails—test accuracy stays near 0%. Each client sees just 24 samples. The gradient signal from each client is so noisy that aggregation cannot compensate.

6 Grokking Success Rates

Key observations.

- **Centralized-reduced is almost uniformly zero.** Only the ($\alpha = 0.5, K = 2$) cell shows 100% success. Every other cell has $\alpha_{\text{eff}} < 0.198$, placing it below the phase boundary. This confirms the Exp 1 finding: centralised grokking has a hard data threshold.
- **FL IID shows 100% success across most of the grid.** All cells with $\alpha \geq 0.3$ grok with 3/3 seeds, regardless of K. Even the extreme $K = 97$ (29 samples per client at $\alpha = 0.3$) succeeds perfectly.
- **Failure is confined to a small corner.** Only ($\alpha = 0.25, K = 50$) with 67% and ($\alpha = 0.25, K = 97$) with 0% show FL failure. The $\alpha = 0.2$ row is a negative control (insufficient total data).
- **The aggregation effect spans a factor of $\sim 50\times$ in per-client data.** From $\alpha_{\text{eff}} = 0.15$ (at $\alpha = 0.3, K = 2$) down to $\alpha_{\text{eff}} = 0.003$ (at $\alpha = 0.3, K = 97$), FL reliably groks. The per-client data varies by $50\times$, yet the outcome is identical.

7 Aggregation Benefit

Key observations.

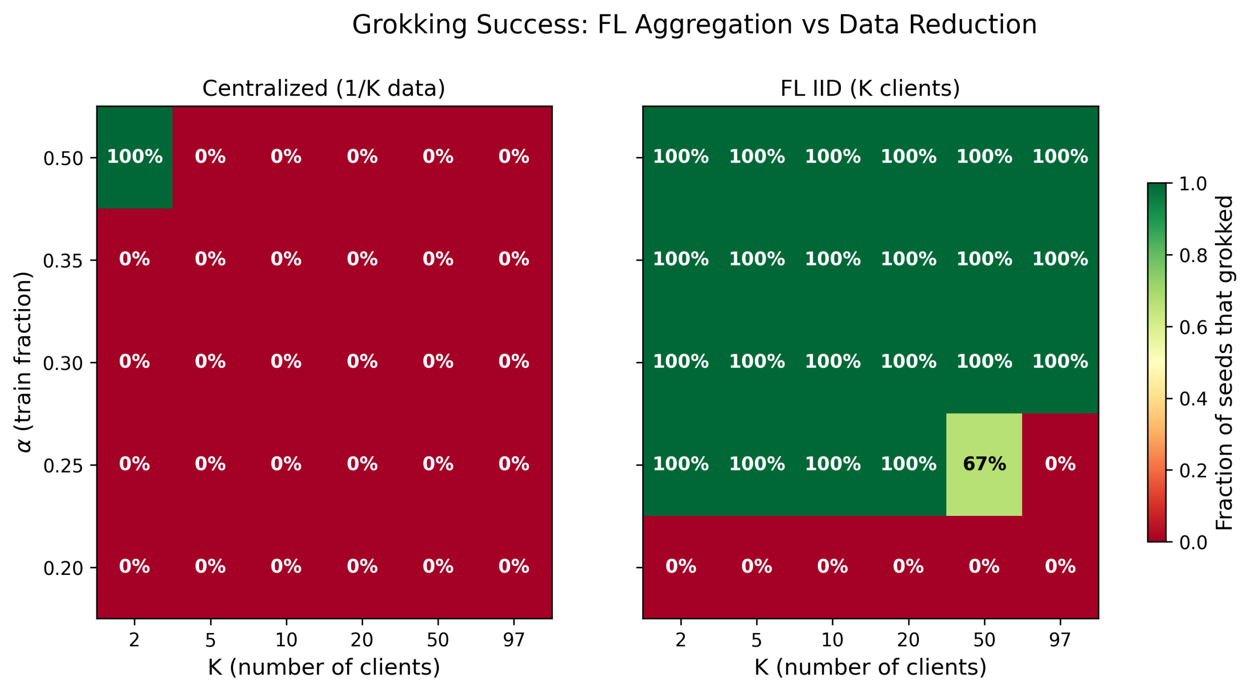

Figure 5: Grokking success rate (fraction of 3 seeds that reach 95% test accuracy). **Left**: centralized-reduced. **Right**: FL IID. The contrast is stark: centralized-reduced groks in only 1 of 30 cells, while FL groks in 23 of 30.

- **The benefit is ∞ almost everywhere.** In 22 of 24 above-boundary cells, FL groks but centralized-reduced does not. Aggregation is not merely helpful—it is *necessary*. Without it, the model on per-client data never generalises.
- **The only finite benefit is** $3.3\times$ at $(\alpha = 0.5, K = 2)$, where centralized-reduced also groks but $3.3\times$ slower than FL. Even in this most favourable case, aggregation provides substantial speedup.
- **No cell shows FL slower than centralized-reduced.** Aggregation never hurts. It either provides infinite benefit (rescuing grokking entirely) or finite speedup.

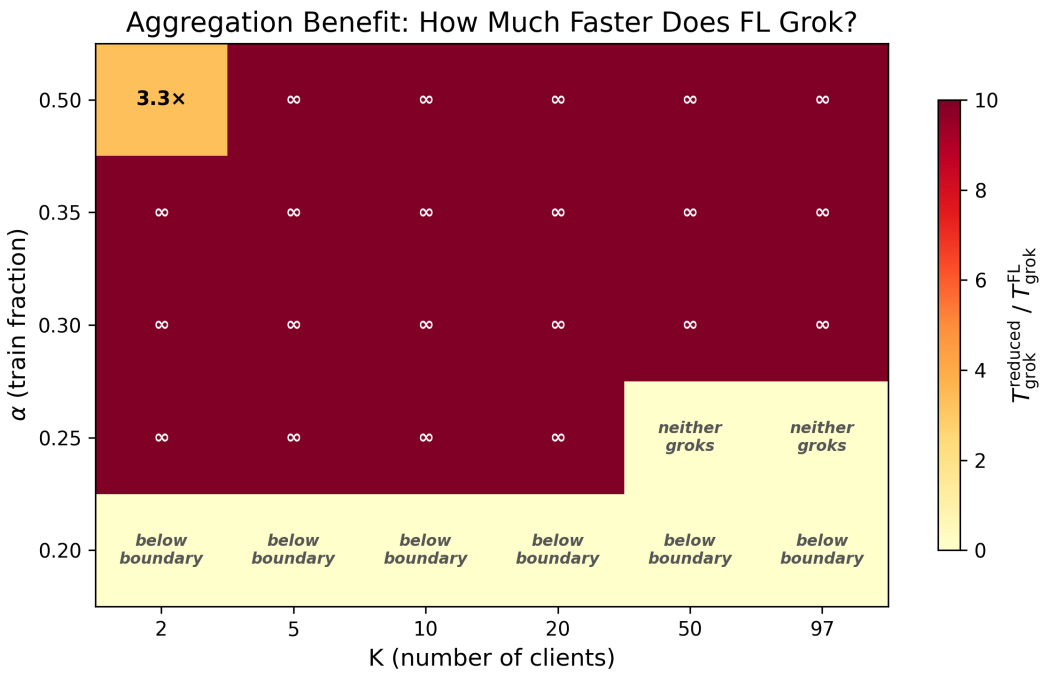

Figure 6: Aggregation benefit: $T_{\text{grok}}^{\text{reduced}}/T_{\text{grok}}^{\text{FL}}$. Values > 1 mean FL groks faster than centralized-reduced. ∞ means FL groks but centralized-reduced never does—the maximum possible benefit.

8 Mechanistic Analysis

8.1 Fourier Structure Emergence

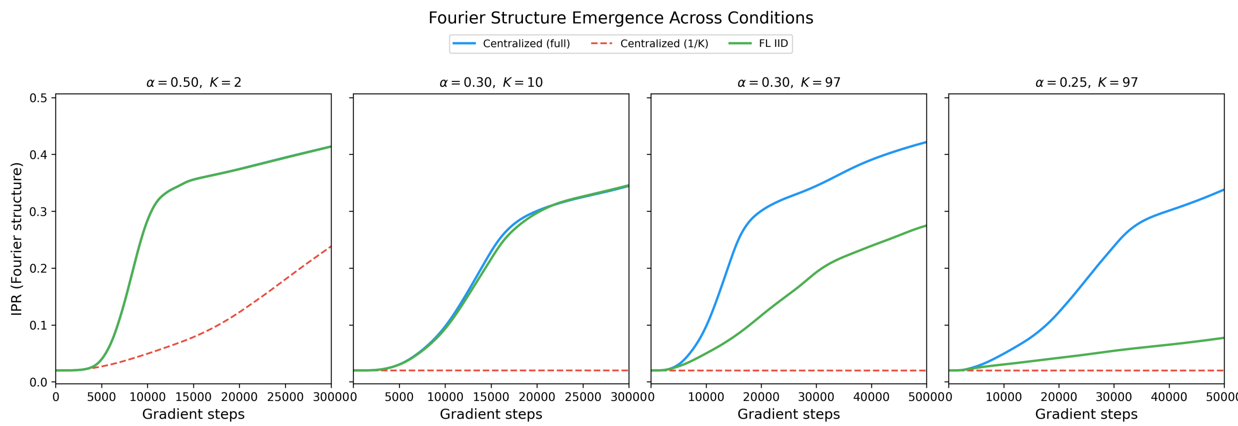

Figure 7: IPR (Fourier structure metric) over time for four representative cells. When grokking occurs, IPR rises from ~ 0.02 (random) to ~ 0.35–0.47 (structured). Centralized-reduced (red dashed) never develops Fourier structure in any panel except the first.

Key observations.

- **FL develops the same Fourier structure as centralised training.** In panels where FL groks, its IPR trajectory closely tracks the centralized-full curve, with only a mild delay at high K.
- **Centralized-reduced shows zero Fourier development** (IPR stays at ~ 0.02) in every panel except ($\alpha = 0.5, K = 2$). The network memorises its tiny dataset but never discovers the periodic structure. This is the mechanistic explanation for why it fails to generalise.
- **At** ($\alpha = 0.25, K = 97$)**, FL also fails to develop Fourier structure.** Its IPR curve stays flat, mirroring the centralized-reduced behaviour. Fragmentation has effectively destroyed the collective gradient signal.

8.2 Same Pathway to Grokking

Key observations.

- **The grokking pathway is universal.** Both centralized and FL trajectories follow the same S-curve in (IPR, test accuracy) space: IPR must reach ~ 0.05–0.10 before test accuracy begins rising, then accuracy and IPR increase together.
- **FL does not take a different route—it takes the same route at a different speed.** This is consistent with the Gromov (2023) picture: grokking is driven by a single Fourier mode becoming dominant, and FedAvg preserves this mechanism.
- **Endpoint IPR is slightly lower for high-K FL runs** (~ 0.35 vs. ~ 0.47 for centralized), suggesting that aggregation noise slightly diffuses the Fourier representation. However, this does not prevent grokking.

8.3 Weight Norm Dynamics

Key observations.

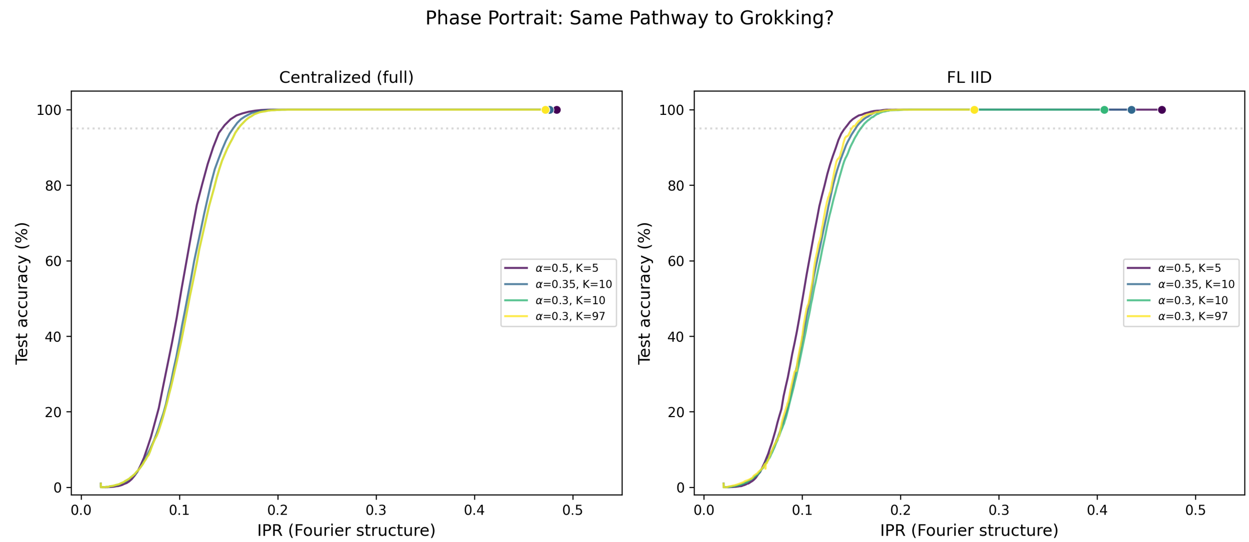

Figure 8: Phase portrait: trajectories in (IPR, test accuracy) space. **Left**: centralized-full. **Right**: FL IID. Multiple (α, K) configurations overlay near-perfectly, indicating a universal grokking pathway.

- **FL weight norms track centralized-full closely**, confirming that FedAvg preserves the global weight dynamics even though local updates are computed on data shards.
- **Centralized-reduced norms plateau early** (red dashed lines flatten while blue/green continue growing). The memorised solution on small data requires less weight magnitude, and the network never transitions to the structured (Fourier) solution that requires larger norms.
- **At extreme fragmentation** ($\alpha = 0.25, K = 97$, panel 4), FL norms grow but more slowly and erratically than centralized-full. The aggregated gradient is noisy, slowing the weight growth needed for the phase transition.

8.4 Loss Dynamics: Memorisation Before Generalisation

Key observations.

- **All conditions memorise quickly.** Train loss drops to $< 10^{-4}$ within $\sim 5{,}000$ steps for centralized-full and FL, regardless of whether grokking eventually occurs. Memorisation is not the bottleneck.
- **Test loss reveals the grokking moment.** For grokking conditions (blue and green), test loss initially rises or plateaus, then plunges—the inflection marks the onset of generalisation. For non-grokking conditions (red dashed), test loss stays flat permanently.
- **FL test loss mirrors centralized-full** in panels (a) and (b), with only a slight delay at high K. In panel (c), FL test loss stays flat (no grokking), behaving like the centralized-reduced baseline.

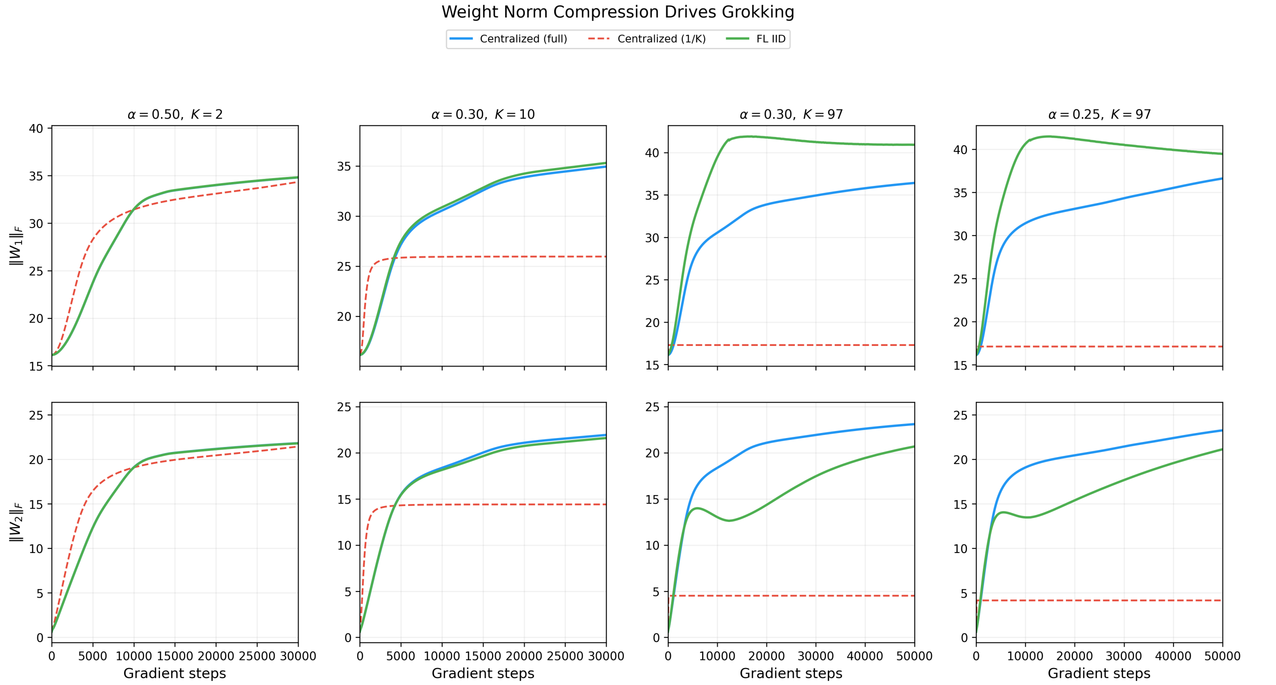

Figure 9: Weight norms ($\|W_1\|_F$ and $\|W_2\|_F$) over time. Grokking coincides with sustained weight growth; failed conditions show early plateauing or divergent growth without structure formation.

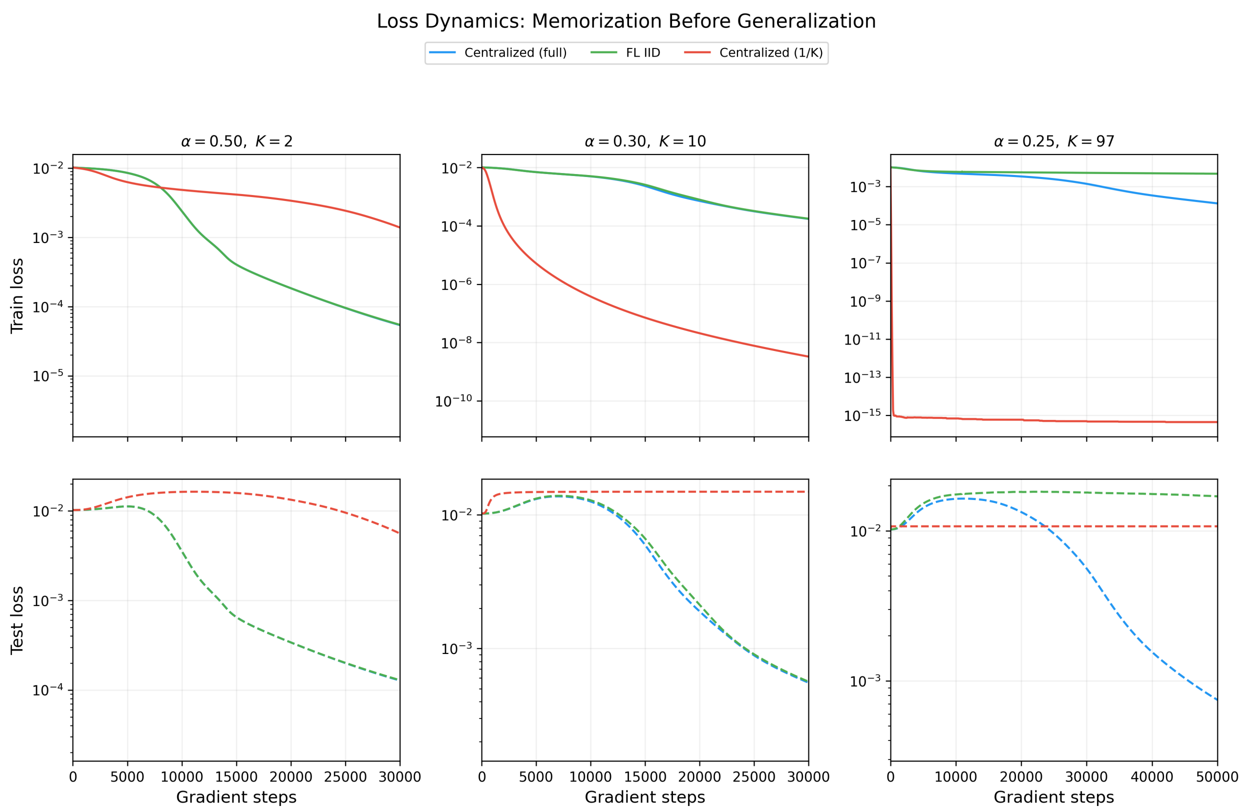

Figure 10: Train loss (top) and test loss (bottom) on log scale. The classic grokking signature: train loss drops immediately (memorisation), test loss follows much later (generalisation).

9 Client Heterogeneity and Drift

9.1 Client Drift vs Number of Clients

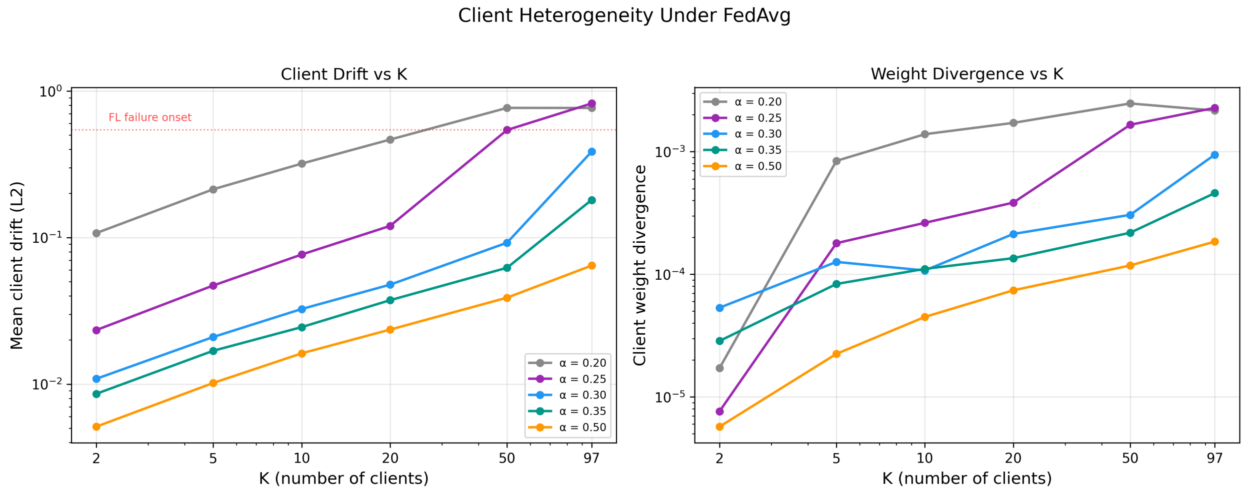

Figure 11: Final client drift (L2 norm of local update minus global update) and weight divergence vs. K, on log-log axes. The red dashed line marks the drift level at which FL begins to fail.

Key observations.

- **Drift increases monotonically with** K as expected: more clients means less data per client, which means more overfitting during local epochs.
- **Higher** α **produces less drift at matched** K**.** With more total data, each client's shard is larger and more representative, reducing local overfitting. The $\alpha = 0.5$ lines are consistently lowest.
- **FL failure correlates with a drift threshold.** The red dashed line marks the drift level of ($\alpha = 0.25, K = 50$), where FL begins failing. Below this threshold, aggregation successfully corrects client drift; above it, the noise overwhelms the signal.

9.2 Drift Dynamics During Grokking

Key observations.

- **Drift peaks precisely at the grokking transition.** In panels (a)–(c), client drift rises during memorisation, peaks near T_{grok}, then drops as the global model converges to the Fourier solution. Once all clients agree on the correct representation, local updates become consistent.
- **The drift peak is sharper at smaller** K**.** At $K = 5$ (panel a), drift spikes and resolves within $\sim 5{,}000$ steps. At $K = 50$ (panel c), the peak is broader and the resolution slower, reflecting the harder coordination problem.
- **At** ($\alpha = 0.25, K = 97$)**, drift grows without bound.** Panel (d) shows drift increasing monotonically with no peak and no accuracy improvement. The system never finds the Fourier attractor—client updates remain permanently incoherent.

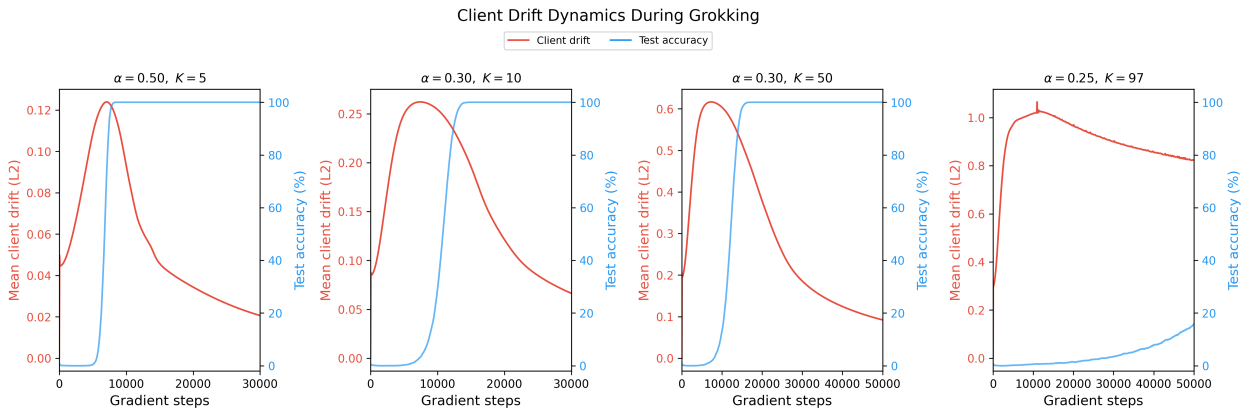

Figure 12: Client drift (red, left axis) and test accuracy (blue, right axis) over time. Drift peaks during the grokking transition and drops sharply once the network settles into the Fourier solution.

10 Reproducibility

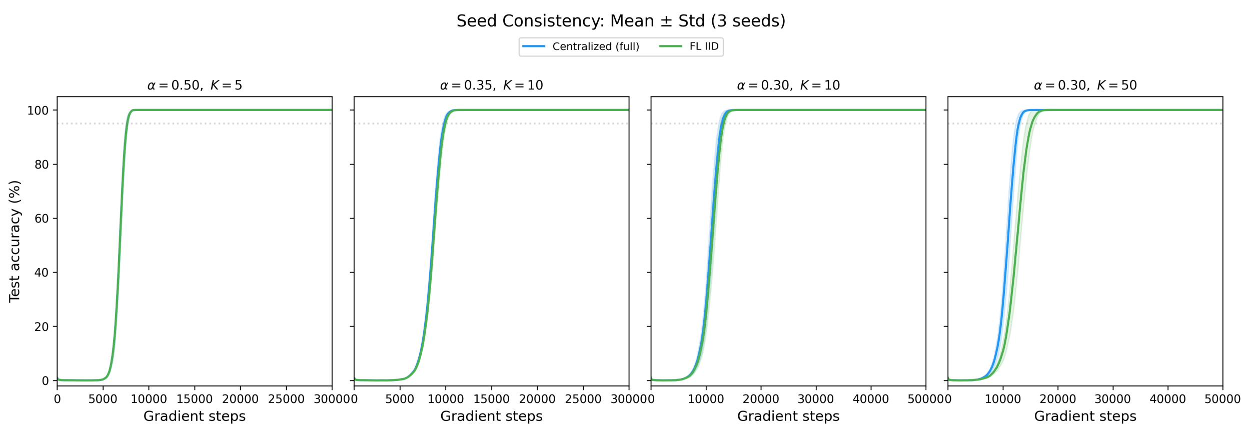

Figure 13: Test accuracy: mean ± std across 3 seeds. Shaded bands show inter-seed variance. The grokking transition is highly reproducible for both centralized and FL.

Key observations.

- **The grokking transition is sharp and reproducible.** At $(\alpha = 0.5, K = 5)$ and $(\alpha = 0.35, K = 10)$, the std bands are nearly invisible—all 3 seeds agree to within a few hundred steps.
- **Variance increases near the phase boundary.** At $(\alpha = 0.3, K = 10)$, FL shows slightly wider bands than centralized during the transition. At $(\alpha = 0.3, K = 50)$, the FL transition is noticeably more spread, though all seeds eventually grok.
- **FL does not introduce qualitative irreproducibility.** Where FL groks, it does so consistently across seeds. The only stochastic failure is at $(\alpha = 0.25, K = 50)$, where 2/3 seeds succeed—right at the fragmentation boundary.

11 Summary Table

Table 1: Experiment 2: FL grokking time and slowdown ratio across the (α, K) grid. Cells where FL fails are marked ∞. Gray rows indicate $\alpha = 0.2$ (below centralised phase boundary—negative control).

α	K	α_{eff}	$T_{\text{grok}}^{\text{cent}}$	$T_{\text{grok}}^{\text{FL}}$	Grokked	Ratio
0.20	2–97	≤ 0.10	∞	∞	0/3	—
0.25	2	0.125	25,133	$25{,}263 \pm 1{,}953$	3/3	1.01×
0.25	5	0.050	25,133	$25{,}783 \pm 2{,}128$	3/3	1.03×
0.25	10	0.025	25,133	$26{,}673 \pm 2{,}172$	3/3	1.06×
0.25	20	0.012	25,133	$29{,}987 \pm 2{,}914$	3/3	1.19×
0.25	50	0.005	25,133	—	2/3	—
0.25	97	0.003	25,133	∞	0/3	—
0.30	2	0.150	12,833	$12{,}822 \pm 449$	3/3	1.00×
0.30	5	0.060	12,833	$12{,}920 \pm 455$	3/3	1.01×
0.30	10	0.030	12,833	$13{,}097 \pm 463$	3/3	1.02×
0.30	20	0.015	12,833	$13{,}480 \pm 492$	3/3	1.05×
0.30	50	0.006	12,833	$15{,}048 \pm 804$	3/3	1.17×
0.30	97	0.003	12,833	$27{,}422 \pm 2{,}655$	3/3	2.14×
0.35	2	0.175	9,867	$9{,}838 \pm 196$	3/3	1.00×
0.35	5	0.070	9,867	$9{,}882 \pm 197$	3/3	1.00×
0.35	10	0.035	9,867	$9{,}952 \pm 190$	3/3	1.01×
0.35	20	0.017	9,867	$10{,}083 \pm 198$	3/3	1.02×
0.35	50	0.007	9,867	$10{,}590 \pm 210$	3/3	1.07×
0.35	97	0.004	9,867	$12{,}837 \pm 414$	3/3	1.30×
0.50	2	0.250	7,667	$7{,}632 \pm 116$	3/3	1.00×
0.50	5	0.100	7,667	$7{,}640 \pm 117$	3/3	1.00×
0.50	10	0.050	7,667	$7{,}642 \pm 120$	3/3	1.00×
0.50	20	0.025	7,667	$7{,}657 \pm 120$	3/3	1.00×
0.50	50	0.010	7,667	$7{,}737 \pm 128$	3/3	1.01×
0.50	97	0.005	7,667	$7{,}945 \pm 152$	3/3	1.04×

12 Conclusions

- **FedAvg aggregation rescues grokking in 23 of 24 above-boundary cells.** In these cells, each client holds data far below the centralised phase boundary ($\alpha_{\text{eff}} \ll \alpha_c$), yet the federated model groks. A centralised model trained on the same per-client data never groks. Aggregation is the sole explanatory variable.
- **The overhead is small.** For $K \leq 20$, the slowdown is $< 1.2\times$ across all α. Even at the extreme $K = 97$, FL groks (for $\alpha \geq 0.3$) with at most a $2.1\times$ slowdown. FedAvg is a surprisingly efficient mechanism for pooling gradient signal.
- **FL preserves the grokking mechanism.** Phase portraits, IPR trajectories, weight norm dynamics, and loss curves all confirm that FL follows the same mechanistic pathway as centralized training. FL does not find a different solution—it finds the same Fourier solution at roughly the same speed.
- **There is a fragmentation limit.** At $\alpha = 0.25$ with $K \geq 50$ (24–47 samples per client), FL begins to fail. The drift analysis shows this corresponds to a critical client drift threshold. Below this threshold, aggregation corrects local overfitting; above it, the signal-to-noise ratio in the aggregated gradient is too low.
- **Client drift peaks at the grokking transition.** This is a novel observation: drift is highest precisely when the model is transitioning from memorisation to the Fourier solution, then drops once all clients converge to the same representation. This suggests grokking itself acts as a natural drift-reduction mechanism.
- $\alpha = 0.2$ **confirms the negative control.** When total data is below the centralized phase boundary, neither centralised nor FL groks at any K. Aggregation cannot create grokking from insufficient total information.

Implications for Experiment 3. The aggregation effect is now established for IID partitions. The natural next question: does it survive *non-IID* data heterogeneity? Exp 3 will test the same grid with operand-based and target-based partitions, where clients see systematically different subsets of the modular arithmetic task.